Packet Tracer - VLAN Configuration

Addressing Table

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	172.17.10.21	255.255.255.0	10
PC2	NIC	172.17.20.22	255.255.255.0	20
PC3	NIC	172.17.30.23	255.255.255.0	30
PC4	NIC	172.17.10.24	255.255.255.0	10
PC5	NIC	172.17.20.25	255.255.255.0	20
PC6	NIC	172.17.30.26	255.255.255.0	30

Objectives

Part 1: Verify the Default VLAN Configuration

Part 2: Configure VLANs

Part 3: Assign VLANs to Ports

Background

VLANs are helpful in the administration of logical groups, allowing members of a group to be easily moved, changed, or added. This activity focuses on creating and naming VLANs, and assigning access ports to specific VLANs.

Part 1: View the Default VLAN Configuration

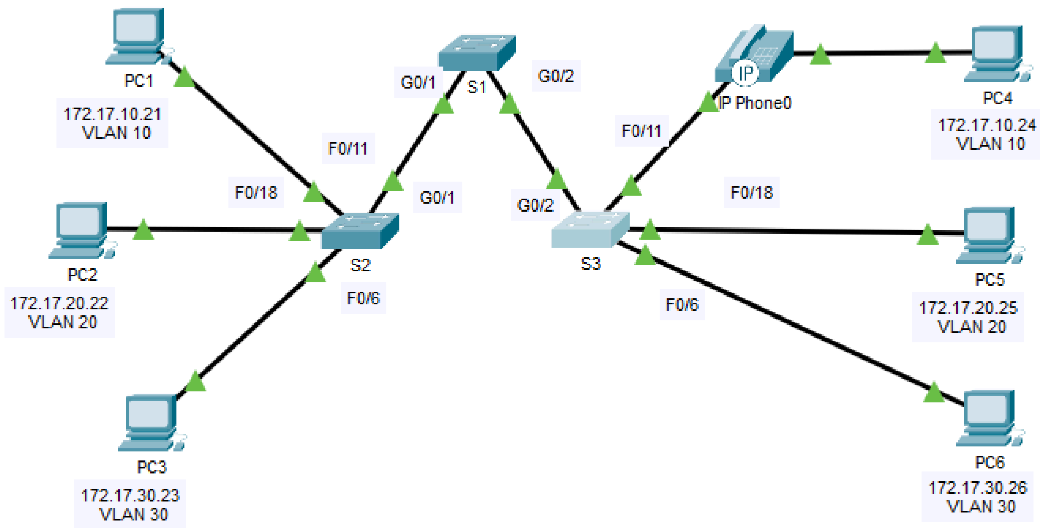
Step 1: Display the current VLANs.

On S1, issue the command that displays all VLANs configured. By default, all interfaces are assigned to VLAN 1.

show vlan brief

Step 2: Verify connectivity between PCs on the same network.

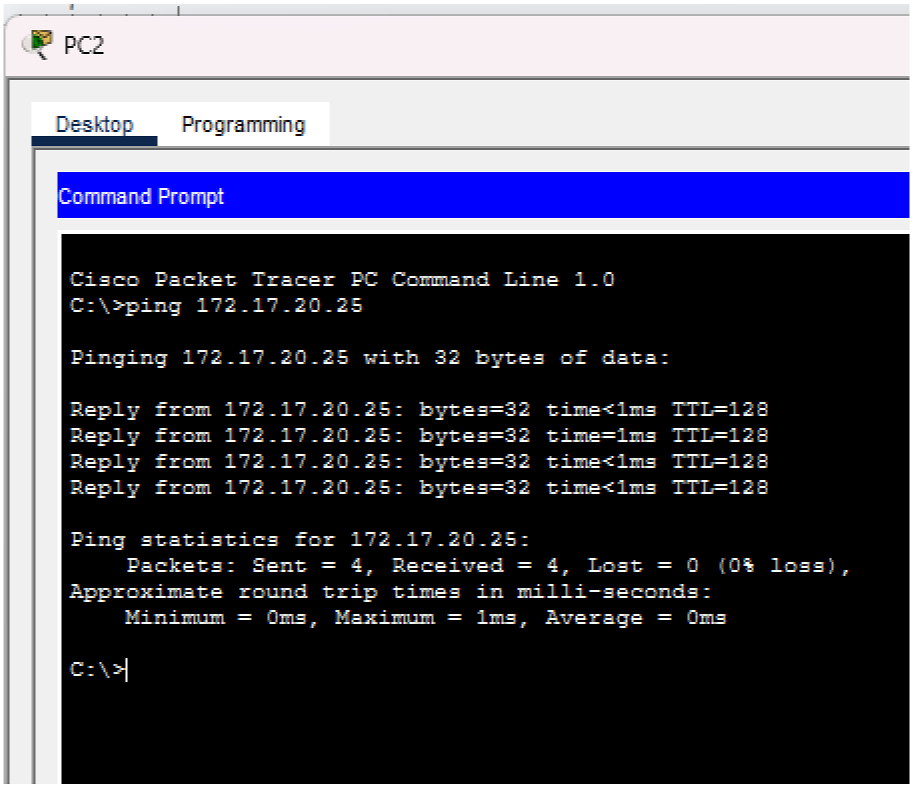
Notice that each PC can ping the other PC that shares the same subnet.



- PC1 can ping PC4

```
Control-C
^C
C:\>ping 172.17.20.22

Pinging 172.17.20.22 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.20.22:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Control-C
^C
C:\>
C:\>ping 172.17.10.24

Pinging 172.17.10.24 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.10.24:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- PC2 can ping PC5



After try:

S2(config)# interface GigabitEthernet 0/1

S2(config-if)# switchport mode trunk

S1(config)# interface range GigabitEthernet 0/1 - 2

S1(config-if)# switchport mode trunk

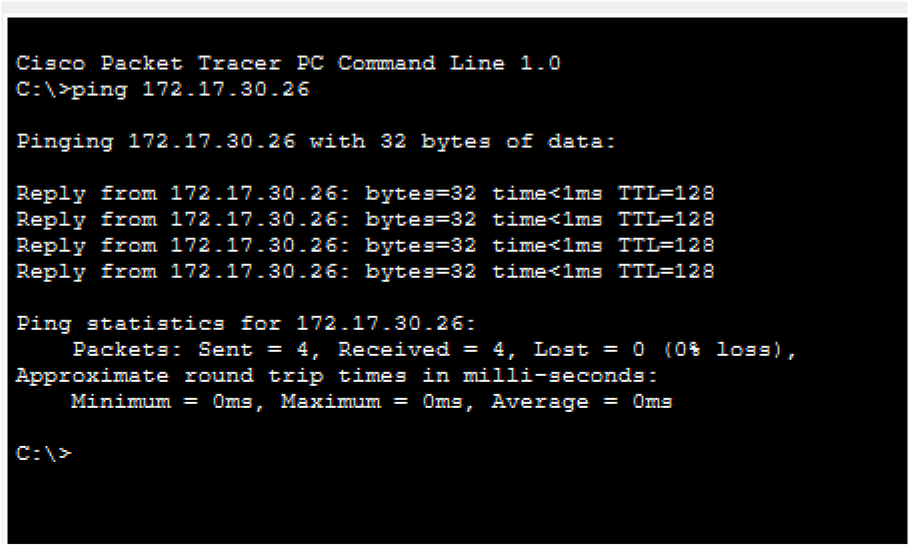
S3(config)# interface GigabitEthernet 0/1

S3(config-if)# switchport mode trunk

Gig0/1 and Gig0/2 (ports connecting switches S2↔S1) are still in VLAN 1 (default) and are not active trunk ports, even after previously typing the command `switchport mode trunk`.

- PC3 can ping PC6

It couldn't ping as well. Like PC1 and PC4



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.17.30.26

Pinging 172.17.30.26 with 32 bytes of data:

Reply from 172.17.30.26: bytes=32 time<1ms TTL=128
Reply from 172.17.30.26: bytes=32 time<1ms TTL=128
Reply from 172.17.30.26: bytes=32 time<1ms TTL=128
Reply from 172.17.30.26: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.30.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Pings to hosts on other networks fail. YES

What benefits can VLANs provide to the network?

Protect the Network about how It can manage who can access It นะคะ

Part 2: Configure VLANs

Step 1: Create and name VLANs on S1.

a. Create the following VLANs. Names are case-sensitive and must match the requirement exactly:

   - VLAN 10: Faculty/Staff

```
S1#(config)# vlan 10
S1#(config-vlan)# name Faculty/Staff
```

b. Create the remaining VLANS.

   - VLAN 20: Students
   - VLAN 30: Guest(Default)
   - VLAN 99: Management&Native
   - VLAN 150: VOICE

```
S1>enable
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#name Faculty/Staff
S1(config-vlan)#exit
S1(config)#enable
% Incomplete command.
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show running configure
                ^
% Invalid input detected at '^' marker.

S1#show running-config
Building configuration...

Current configuration : 1082 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
```

```
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#vlan 20
S1(config-vlan)#name Students
S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name Guest(Default)
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 99
S1(config-vlan)#name Management&Native
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 150
S1(config-vlan)#name VOICE
S1(config-vlan)#
```

Step 2: Verify the VLAN configuration.

Which command will only display the VLAN name, status, and associated ports on a switch?

show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Faculty/Staff                    active    Fa0/11
20   Students                         active    Fa0/18
30   Guest(Default)                   active    Fa0/6
99   Management&Native                active
150  VOICE                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

Step 3: Create the VLANs on S2 and S3.

Use the same commands from Step 1 to create and name the same VLANs on S2 and S3.

Step 4: Verify the VLAN configuration.

```
S2>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Faculty/Staff                    active    Fa0/11
20   Students                         active    Fa0/18
30   Guest(Default)                   active    Fa0/6
99   Management&Native                active
150  VOICE                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S2>
```

```
S3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Faculty/Staff                    active    Fa0/11
20   Students                         active    Fa0/18
30   Guest(Default)                   active    Fa0/6
99   Management&Native                active
150  VOICE                            active    Fa0/11
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S3#
```

Part 3: Assign VLANs to Ports

Step 1: Assign VLANs to the active ports on S2.

a. Configure the interfaces as access ports and assign the VLANs as follows:

  - VLAN 10: FastEthernet 0/11

S2(config)# **interface f0/11**

S2(config-if)# **switchport mode access**

S2(config-if)# **switchport access vlan 10**

b. Assign the remaining ports to the appropriate VLAN.

  - VLAN 20: FastEthernet 0/18
  - VLAN 30: FastEthernet 0/6

```
S2>enable
S2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#vlan 10
S2(config-vlan)#name Faculty/Staff
S2(config-vlan)#vlan20
                  ^
% Invalid input detected at '^' marker.

S2(config-vlan)#vlan 20
S2(config-vlan)#name Students
S2(config-vlan)#vlan 30
S2(config-vlan)#name Guest(Default)
S2(config-vlan)#vlan 99
S2(config-vlan)#name Management&Native
S2(config-vlan)#vlan 150
S2(config-vlan)#name VOICE
S2(config-vlan)#no shutdown
                   ^
% Invalid input detected at '^' marker.

S2(config-vlan)#exit
S2(config)#noshutdown
              ^
% Invalid input detected at '^' marker.

S2(config)#no shutdown
               ^
% Invalid input detected at '^' marker.

S2(config)#
S2(config)#interface FastEthernet0/11
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 10
S2(config-if)#exit
S2(config)#interface FastEthernet0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan20
                                   ^
% Invalid input detected at '^' marker.

S2(config-if)#switchport access vlan 20
S2(config-if)#interface FastEthernet0/6
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 30
S2(config-if)#exit
S2(config)#
```

Step 2: Assign VLANs to the active ports on S3.

S3 uses the same VLAN access port assignments as S2. Configure the interfaces as access ports and assign the VLANs as follows:

- VLAN 10: FastEthernet 0/11
- VLAN 20: FastEthernet 0/18
- VLAN 30: FastEthernet 0/6

Step 3: Assign the VOICE VLAN to FastEthernet 0/11 on S3.

As shown in the topology, the S3 FastEthernet 0/11 interface connects to a Cisco IP Phone and PC4. The IP phone contains an integrated three-port 10/100 switch. One port on the phone is labeled Switch and connects to F0/4. Another port on the phone is labeled PC and connects to PC4. The IP phone also has an internal port that connects to the IP phone functions.

The S3 F0/11 interface must be configured to support user traffic to PC4 using VLAN 10 and voice traffic to the IP phone using VLAN 150. The interface must also enable QoS and trust the Class of Service (CoS) values assigned by the IP phone. IP voice traffic requires a minimum amount of throughput to support acceptable voice communication quality. This command helps the switchport to provide this minimum amount of throughput.

S3(config)# **interface f0/11**

S3(config-if)# **mls qos trust cos**

S3(config-if)# **switchport voice vlan 150**

```
S3>enable
S3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S3(config)#vlan 10
S3(config-vlan)#name Faculty/Staff
S3(config-vlan)#vlan 20
S3(config-vlan)#name Students
S3(config-vlan)#vlan 30
S3(config-vlan)#name Guest(Default)
S3(config-vlan)#vlan 99
S3(config-vlan)#name Management&Native
S3(config-vlan)#vlan150
                ^
% Invalid input detected at '^' marker.

S3(config-vlan)#vlan 150
S3(config-vlan)#name VOICE
S3(config-vlan)#exit
S3(config)#
S3(config)#interface FastEthernet0/18
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan20
                                    ^
% Invalid input detected at '^' marker.

S3(config-if)#switchport access vlan 20
S3(config-if)#exit
S3(config)#interface FastEthernet0/6
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 30
S3(config-if)#exit
S3(config)#interface FastEthernet0/11
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 10
S3(config-if)#mls qos trust cos
S3(config-if)#switchport voice vlan 150
S3(config-if)#exit
S3(config)#
```

Step 4: Verify loss of connectivity.

Previously, PCs that shared the same network could ping each other successfully.

Study the output of from the following command on **S2** and answer the following questions based on your knowledge of communication between VLANS. Pay close attention to the Gig0/1 port assignment.

S2# **show vlan brief**

```
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/19
```

```
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 Faculty/Staff active Fa0/11
20 Students active Fa0/18
30 Guest(Default) active Fa0/6
99 Management&Native active
150 VOICE active
```

Try pinging between PC1 and PC4.

Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

No. PC1 fails to ping PC4 even though both machines are in the same VLAN 10 (172.17.10.x) because the ports connecting the switches (S2 Gig0/1 → S1 and S1 → S3) are still classified in VLAN 1 (default) as shown in the show vlan brief, and are not trunk ports that tag and pass VLAN 10 traffic across the switches. Therefore, frames from PC1 (VLAN 10) cannot travel across S2 to S1 and S3 to reach PC4. — Each VLAN traffic is confined within a single switch if the link between switches is not a trunk.

What could be done to resolve this issue?

You need to check and configure the ports connected between the switches (S1–S2, S1–S3) as trunk ports correctly, and then confirm with the `show interfaces trunk` command (not just `show vlan brief`) to see if the ports have actually become trunking.

Commands to rerun/check:

S2(config)# interface GigabitEthernet0/1

S2(config-if)# switchport mode trunk

S2(config-if)# switchport trunk allowed vlan all

S2(config-if)# end

S2# show interfaces trunk